piled one on top of another. Making all that go away will take some doing.

Imagine bombarding Venus with asteroids and comets. Each impact would blow away some of the atmosphere. To blow away almost all of it, though, would require using up more big asteroids and comets than there are—at least in the planetary part of the Solar System. Even if that many potential impactors existed, even if we could make them all collide with Venus (this is the overkill approach to the impact hazard problem), think what we would have lost. Who knows what wonders, what practical knowledge they contain? We would also obliterate much of Venus' gorgeous surface geology—which we've just begun to understand, and which may teach us much about the Earth. This is an example of brute-force terraforming. I suggest we want to steer entirely clear of such methods, even if someday we'll be able to afford them (which I very much doubt). We want something more elegant, more subtle, more respectful of the environments of other worlds. A microbial approach has some of those virtues, but does not do the trick, as we've just seen.

We can imagine pulverizing a dark asteroid and spreading the powder through the upper atmosphere of Venus, or carrying such dust up from the surface. This would be the physical equivalent of nuclear winter or the Cretaceous-Tertiary post-impact climate. If the sunlight reaching the ground is sufficiently attenuated, the surface temperature must fall. But by its very nature, this option plunges Venus into deep gloom, with daytime light levels perhaps only as bright as on a moonlit night on Earth. The oppressive, crushing 90-bar atmosphere would remain untouched. Since the emplaced dust would sediment out every few years, the layer would have to be replenished in the same period of time. Perhaps such an approach would be acceptable for short exploratory missions, but the environment generated seems very stark for a self-sustaining human community on Venus.

We could use a giant artificial sunshade in orbit around Venus to cool the surface; but it would be enormously expensive, as well as having many of the deficiencies of the dust layer. However, if the